The Complete Treatise on Structural Identification of Sovereign Bond Pricing with a Three-Class K-Means Classifier

From Continuous-Time Bond Valuation to an Endogenous Classification of Sovereign Risk

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Abstract

This treatise develops a structural and empirical framework for studying sovereign bond prices without assuming in advance that any sovereign issuer is absolutely risk-free. It combines continuous-time discounted-cash-flow valuation, duration analysis, latent common-component decomposition, econometric identification, graph-based regularization, and an unsupervised three-cluster K-Means procedure. The intended economic labels are *Safe*, *Risky*, and *Default*; these labels are not native outputs of K-Means and require an explicit, independently validated interpretation rule. The framework distinguishes market convention, issuer representation, regulatory treatment, relative safety, creditworthiness, and realised default. It develops identification propositions, derives estimators, specifies a reproducible algorithm, and proposes empirical tests, data governance, cyber-resilience, and decision-use safeguards. Connections to law, political science, geopolitics, diplomacy, international relations, welfare economics, warfare economics, physics, chemistry, biology, linguistics, psychology, psychiatry, philosophy, and religion are treated as analytical perspectives rather than substitutes for financial evidence. The principal conclusion is methodological: a bond collection can act as a structured classifier of relative sovereign risk, but neither a statistical cluster nor a common yield component proves that an issuer is risk-free.

The treatise ends with "The End"

Keywords: sovereign bonds; term structure; structural identification; common components; K-Means; default risk; safe assets; econometrics; systemic risk; financial stability; model governance.

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1 Introduction

Sovereign debt occupies several roles at once [8, 12]. It is a claim on a public issuer, a financing instrument, a benchmark for other assets, collateral in financial transactions, a component of institutional portfolios, and a political commitment whose credibility depends on fiscal capacity and institutional arrangements. These roles need not coincide. A sovereign bond can be liquid without being credit-risk-free, highly rated without being immune to inflation or currency risk, or treated as a benchmark while carrying material duration and market risk.

This motivates a structural question: can the joint pricing structure of a collection of sovereign bonds help identify relative risk and classify issuers without declaring one sovereign to be risk-free by assumption? The proposed answer is conditional. Joint data can reveal relative differences and latent common movements, but absolute economic interpretations require identifying restrictions, external evidence, and validation.

The framework has two linked layers. The first estimates economically interpretable pricing components from bond observations. The second maps a feature vector built from those estimates and independent indicators into three clusters using K-Means. A subsequent interpretation rule may assign the names *Safe*, *Risky*, and *Default*, provided that the empirical evidence supports those names.

1.1 Research questions and epistemic discipline

The treatise asks: how should continuous-time prices and duration be represented; which pricing components are identified; how can a common component be estimated; how can K-Means be integrated without confusing clustering with default prediction; and how should the system be validated for economic, legal, and operational use? The mathematical discussion distinguishes identities, propositions conditional on assumptions, and empirical hypotheses. A regression decomposition is not proof that a latent component is risk-free. A cluster is not a legal finding of default. These distinctions prevent a useful analytical architecture from becoming an overconfident label generator.

2 Continuous-time bond valuation

2.1 Deterministic discounting

Let a bond be observed at time t, mature at T, and pay a coupon flow $C(s)$ at time s and terminal payment $M(T)$. Let $r(u)$ be a deterministic instantaneous discount rate. Define the discount factor

$$Z(t,s) = \exp\left(-\int_t^s r(u)\,du\right). \tag{1}$$

The discounted-cash-flow price is

$$P(t) = \int_t^T C(s)Z(t,s)\,ds + M(T)Z(t,T). \tag{2}$$

This representation assumes known cash flows and an appropriate discount curve. For a risky sovereign claim, a deterministic risk-free discount curve is not automatically justified; the formula is conditional on the selected discounting convention.

Since $\partial Z(t,s)/\partial t = r(t)Z(t,s)$, Leibniz's rule gives

$$\begin{aligned} P'(t) &= -C(t) + \int_t^T C(s)\frac{\partial Z(t,s)}{\partial t}\,ds + M(T)\frac{\partial Z(t,T)}{\partial t} && (3)\\ &= -C(t) + r(t)\left[\int_t^T C(s)Z(t,s)\,ds + M(T)Z(t,T)\right] && (4)\\ &= r(t)P(t) - C(t), && (5) \end{aligned}$$

with terminal condition $P(T) = M(T)$. This differential equation restates the deterministic discounted-cash-flow identity.

2.2 Duration as a parallel-shift sensitivity

Shift the remaining discount curve by a constant ϵ:

$$Z_\epsilon(t,s) = Z(t,s)\exp\big(-\epsilon(s-t)\big). \tag{6}$$

Let $P(t;\epsilon)$ denote the corresponding price. Define duration under this parallel-shift convention by

$$D(t) = -\left.\frac{\partial \log P(t;\epsilon)}{\partial \epsilon}\right|_{\epsilon=0}. \tag{7}$$

Differentiating yields

$$D(t) = \frac{\int_t^T (s-t)C(s)Z(t,s)\,ds + (T-t)M(T)Z(t,T)}{P(t)}. \tag{8}$$

Duration is a price-weighted average of remaining cash-flow times under this convention. It is not a complete measure of credit, liquidity, inflation, currency, settlement, or market risk [4].

Define $H(t) = P(t)D(t)$. Then

$$H(t) = \int_t^T (s-t)C(s)Z(t,s)\,ds + (T-t)M(T)Z(t,T), \tag{9}$$

and differentiation gives $H'(t) = P(t) - r(t)H(t)$. Since $D = H/P$ and $P' = rP - C$,

$$D'(t) = \frac{H'(t)P(t) - H(t)P'(t)}{P(t)^2} \tag{10}$$

$$= \frac{[P(t) - r(t)H(t)]P(t) - H(t)[r(t)P(t) - C(t)]}{P(t)^2} \tag{11}$$

$$= 1 - \frac{C(t)}{P(t)}D(t), \tag{12}$$

with $D(T) = 0$. This result uses a continuous coupon-flow convention; discrete coupons require a corresponding cash-flow sum.

2.3 Dimensional consistency

A term such as $1 + D(t)$ is dimensionally invalid if $D(t)$ is measured in units of time. Introduce a time scale τ_0 and define

$$\widetilde{D}(t) = \frac{D(t)}{\tau_0}, \tag{13}$$

so that $1 + \widetilde{D}(t)$ is dimensionless. Alternatively, one could use a dimensionally coherent transformation such as $\exp(\kappa D(t))$, where κ has inverse-time units. A model can fit numerically while remaining dimensionally ill-defined, so this issue must be resolved before estimation.

3 A proposed pricing relation and its limits

Suppose an empirical construction defines

$$Q_i(t) = \frac{P_i(t)\left[1 + \widetilde{D}_i(t)\right]}{1 + r_f(t) + p_i(t)}. \tag{14}$$

Assuming all quantities are dimensionally and numerically compatible, algebra gives

$$r_f(t) + p_i(t) = \frac{P_i(t)\left[1 + \widetilde{D}_i(t)\right]}{Q_i(t)} - 1. \tag{15}$$

This identifies the sum on the left conditional on the proposed relation; it does not separately identify r_f and p_i. The symbol $p_i(t)$ should therefore be called an issuer-specific residual or pricing component until a stronger interpretation is established. It is not automatically a credit spread, expected loss, or excess-return premium. Such interpretations require additional assumptions about default, recovery, risk aversion, liquidity, currency, inflation, and the pricing kernel (Cochrane, 2005; Duffie and Singleton, 1999).

4 Structural identification and common-component ambiguity

4.1 Baseline decomposition

Let $\rho_i(t)$ denote an observed or estimated effective pricing rate for sovereign i. Consider

$$\rho_i(t) = g(t) + s_i(t), \tag{16}$$

where $g(t)$ is a common component and $s_i(t)$ is an issuer-specific deviation. For any function $a(t)$, define

$$\widetilde{g}(t) = g(t) + a(t), \tag{17}$$
$$\widetilde{s}_i(t) = s_i(t) - a(t). \tag{18}$$

Then $\widetilde{g}(t) + \widetilde{s}_i(t) = \rho_i(t)$. Thus the observed data alone cannot distinguish every possible common component from every possible issuer-specific component.

4.2 Proposition 1: relative-effect identification

Proposition. Under the additive decomposition, pairwise differences identify relative issuer effects:

$$\rho_i(t) - \rho_j(t) = s_i(t) - s_j(t). \tag{19}$$

They do not, by themselves, identify the absolute level of $g(t)$ or the absolute levels of $s_i(t)$.

Proof. Subtract the equation for issuer j from that for issuer i:

$$\rho_i(t) - \rho_j(t) = [g(t) + s_i(t)] - [g(t) + s_j(t)] \tag{20}$$
$$= s_i(t) - s_j(t). \tag{21}$$

The common component cancels. For any $a(t)$, the transformed components above produce the same observed ρ_i. Hence the absolute decomposition is not unique absent a normalization or additional information. □

A possible normalization is $\sum_i w_i s_i(t) = 0$, where nonnegative weights sum to one. This selects a reference level; it does not prove that the weighted-average issuer is risk-free. Alternative restrictions may use an external benchmark, an instrument, an economic model, or independent risk measures. Each restriction requires economic justification and robustness checks [15, 1].

4.3 Richer structural model and estimator

For N sovereigns, write

$$\boldsymbol{\rho}(t) = \mathbf{1}g(t) + Z(t)\boldsymbol{\beta}(t) + \boldsymbol{\varepsilon}(t). \tag{22}$$

If Z has full column rank and $\mathbf{1}$ is not in its column space, the common intercept is separately estimable within this linear specification. Define

$$M_Z = I - Z(Z^\mathsf{T}Z)^{-1}Z^\mathsf{T}. \tag{23}$$

The least-squares estimate of the common component is

$$\widehat{g}(t) = \frac{\mathbf{1}^\mathsf{T}M_Z\boldsymbol{\rho}(t)}{\mathbf{1}^\mathsf{T}M_Z\mathbf{1}}, \tag{24}$$

and the coefficient estimate is

$$\widehat{\boldsymbol{\beta}}(t) = (Z^\mathsf{T}Z)^{-1}Z^\mathsf{T}\left[\boldsymbol{\rho}(t) - \mathbf{1}\widehat{g}(t)\right]. \tag{25}$$

These formulas follow from the Frisch–Waugh–Lovell partial-regression result and least-squares normal equations [1]. They require invertible matrices and a suitable specification. A small value of $\mathbf{1}^\mathsf{T}M_Z\mathbf{1}$ signals weak separation between the common intercept and the regressors.

4.4 Proposition 2: statistical identification is not risk-freedom

Proposition. Unique estimation of $g(t)$ under a statistical specification does not establish that $g(t)$ is a risk-free rate.

Argument. The estimated common component may reflect monetary policy, shared inflation expectations, global risk appetite, common liquidity premia, collateral convenience, or systematic risk compensation. Statistical uniqueness concerns the mapping from observed data to parameters under assumptions; economic risk-freedom concerns the payoff and risk properties of a specified claim in a specified currency and horizon. These are different propositions. Independent economic interpretation and evidence are required. □

5 Feature engineering for sovereign classification

A candidate feature vector is

$$\mathbf{x}_i(t) = \begin{bmatrix} \widehat{s}_i(t) & \widehat{\sigma}_i(t) & \widehat{d}_i(t) & \widehat{q}_i(t) & \widehat{\lambda}_i(t) & \widehat{m}_i(t) & \widehat{e}_i(t) \end{bmatrix}^{\mathsf{T}}, \tag{26}$$

where the entries may represent:

$\widehat{s}_i$ estimated issuer-specific pricing deviation;

$\widehat{\sigma}_i$ yield, spread, or residual volatility;

$\widehat{d}_i$ independently estimated default-risk indicator;

$\widehat{q}_i$ fiscal or debt-service stress indicator;

$\widehat{\lambda}_i$ liquidity-risk indicator;

$\widehat{m}_i$ sensitivity to global or market stress;

$\widehat{e}_i$ external vulnerability, such as foreign-currency funding exposure.

The list is illustrative. Feature selection should reflect intended use, data availability, measurement quality, and legal constraints. Redundant variables can implicitly receive excess weight; features on larger numerical scales can dominate unless transformed and standardized.

Comparisons across currencies require particular caution. Nominal yields incorporate expected inflation, policy rates, term premia, currency-risk compensation, liquidity, and potentially capital controls. A raw cross-currency yield difference is not a pure sovereign-credit spread. Comparisons should use matched maturities and explicit currency-specific models, and disclose what remains unidentified. A sovereign may be relatively low risk for local-currency nominal repayment but exposed to inflation risk; foreign-currency debt creates a different constraint when the sovereign does not issue the settlement currency. Every classification must state the claim, currency, horizon, and risk definition.

6 Three-class K-Means classifier

6.1 Standardization

For feature j, let x_{ij} denote the value for sovereign i. Conventional standardization is

$$\widetilde{x}_{ij} = \frac{x_{ij} - \overline{x}_j}{s_j}, \tag{27}$$

where $\overline{x}_j$ and s_j are the training-sample mean and standard deviation. A zero-variance feature must be removed or handled explicitly. Robust scaling may be preferable for heavy-tailed variables. Preprocessing parameters must be fitted on training data only and then applied unchanged to validation and test data.

6.2 K-Means objective

For standardized vectors $\widetilde{\mathbf{x}}_i$, K-Means with $K = 3$ minimizes

$$\min_{\mathcal{C}_1,\mathcal{C}_2,\mathcal{C}_3} \sum_{k=1}^{3} \sum_{i \in \mathcal{C}_k} \|\widetilde{\mathbf{x}}_i - \boldsymbol{\mu}_k\|_2^2 , \tag{28}$$

where $\mathcal{C}_k$ is cluster k, and its centroid is

$$\boldsymbol{\mu}_k = \frac{1}{|\mathcal{C}_k|} \sum_{i \in \mathcal{C}_k} \widetilde{\mathbf{x}}_i. \tag{29}$$

The assignment step is

$$\widehat{k}_i = \arg\min_{k \in \{1,2,3\}} \|\widetilde{\mathbf{x}}_i - \boldsymbol{\mu}_k\|_2^2 . \tag{30}$$

The centroid-update step recomputes the mean of each assigned cluster. Iteration reduces the objective until assignments stabilize or a stopping criterion is reached. Since the objective is non-convex, initialization can affect the local solution; multiple initializations are recommended.

6.3 Cluster identifiers versus economic labels

K-Means returns arbitrary identifiers such as 1, 2, and 3. Permuting those identifiers leaves the partition unchanged. The algorithm does not know the meanings of Safe, Risky, or Default. An interpretation map

$$\mathcal{L} : \{1, 2, 3\} \longrightarrow \{\text{Safe}, \text{Risky}, \text{Default}\} \tag{31}$$

must be established after examining the clusters and independent evidence.

A defensible procedure is to fit K-Means using information available at the classification date; inspect centroids, distributions, and stability; compare clusters with independent indicators and documented default events; and map clusters to labels only if their profiles support the intended meanings. If no cluster corresponds coherently to default, do not force one to carry that label merely because $K = 3$. A separate event-based default rule may be necessary. In that case, the architecture is hybrid: clustering characterizes structural similarity, while independently verified events determine Default status.

The Safe label should mean *relatively low measured risk under a stated definition, sample, currency, and horizon.* It is not a universal assertion of zero default, inflation, currency, liquidity, settlement, or market risk.

7 Graph structure and network regularization

Sovereigns are not independent observations. They may share monetary unions, trade links, defence arrangements, creditor bases, regional institutions, or common exposure to global shocks. Let $W_{ij} \geq 0$ denote a predeclared relationship weight. A graph penalty can regularize residuals:

$$\mathcal{L}_{\text{graph}} = \sum_i w_i(\rho_i - g - z_i^{\mathsf{T}}\beta)^2 + \lambda \sum_{i,j} W_{ij}(u_i - u_j)^2 + \mathcal{R}(\beta), \tag{32}$$

where u_i is a residual component and $\mathcal{R}$ is a regularizer. This penalty encourages related nodes to have similar residuals when the graph warrants that assumption. It can also propagate errors, encode political assumptions, or smooth away real differences. The graph must be documented, sensitivity-tested, and constructed without future-information leakage. It does not resolve common-component ambiguity by itself [7, 15].

8 End-to-end architecture

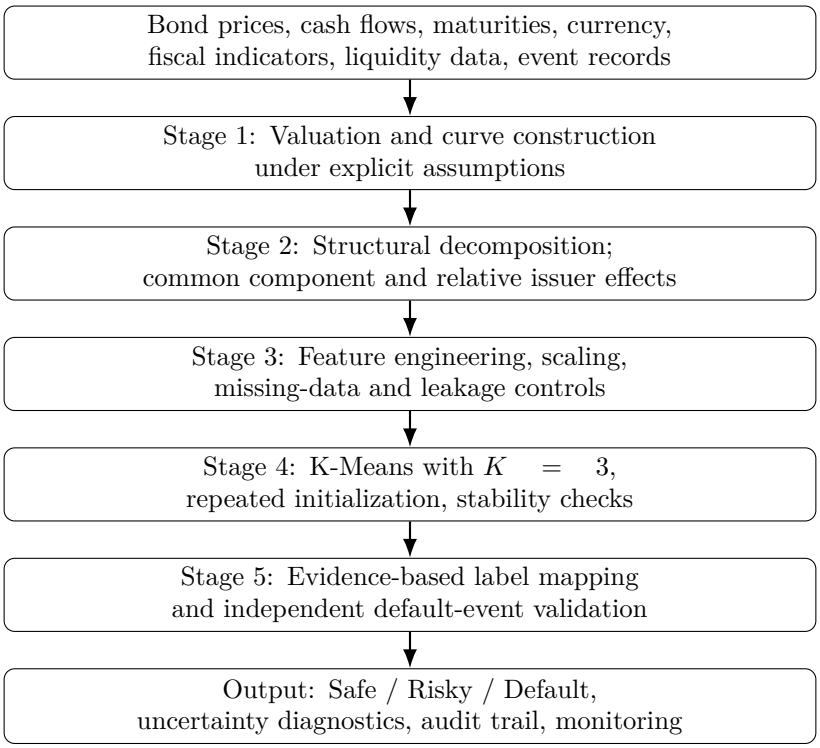


Figure 1: End-to-end architecture.

9 Algorithm and pseudocode

The following workflow separates cluster discovery from label assignment.

```
INPUT:
  Bond observations with timestamps, cash flows, prices, currencies,
  maturities, sovereign identifiers, fiscal/liquidity indicators,
  and independently documented default events.

1. Validate provenance, timestamps, units, currency, and revisions.
2. Construct comparable pricing observations and term-structure features.
3. Estimate the structural model under a declared identification strategy.
4. Store estimates, uncertainty, assumptions, and sensitivity diagnostics.
5. Build features using only information available at time t.
6. Split data by time (and by sovereign where appropriate) before fitting.
7. Fit preprocessing transformations on training data only.
8. Fit K-Means with K = 3 and multiple initializations.
9. Evaluate stability, separation, cluster size, and economic coherence.
10. Compare clusters with independent default-event records and risk measures.
11. Map cluster IDs to Safe, Risky, Default only when justified.
12. If no cluster validly represents Default, use a documented event rule
    or report that the three-label mapping is unsupported.
13. Evaluate on untouched later data; report event metrics and calibration
    for any separate probability model.
14. Save data versions, features, transformations, centroids, label rule,
    validation results, model version, and monitoring thresholds.

OUTPUT:
  Classification, diagnostics, uncertainty warnings, provenance, limitations.
```

10 Econometrics and empirical identification

A useful panel may index sovereign i, bond b, currency c, maturity τ, and time t. A schematic model is

$$\rho_{ibc\tau t} = g_{ct\tau} + s_{it} + \ell_{ibct} + \pi_{ct\tau} + \theta_{ct\tau} \tag{33}$$

$$+ \mathbf{z}_{it}^{\mathsf{T}}\beta + \varepsilon_{ibc\tau t}. \tag{34}$$

This is a decomposition template, not a guarantee that every term is separately estimable. Identification depends on panel support: multiple maturities, repeated observations, currency overlap, explanatory-variable variation, and restrictions on component evolution. Potential methods include fixed effects, factor models, state-space models, Bayesian hierarchical models, and regularized regression [5, 9]. Each method makes assumptions about noise, persistence, cross-sectional dependence, and prior information. Estimation uncertainty should be propagated into the classifier.

Important threats include look-ahead bias, survivorship bias, endogeneity, measurement error, sample selection, regime change, and cross-sectional dependence. Defaults and restructurings must not disappear from the sample simply because their bonds cease trading. Yield data, event data, and fiscal variables should be point-in-time aligned.

Validation should be time-aware. Randomly splitting individual rows can leak information from the same sovereign and adjacent periods into both training and test sets. Prefer rolling-origin or expanding-window evaluation, supplemented by leave-one-sovereign-out tests if deployment includes new issuers. Diagnostics include within-cluster sum of squares, silhouette scores, centroid stability, cluster-size sensitivity, and sensitivity to scaling and feature choices. A high silhouette score establishes geometric separation under a chosen metric, not economic correctness.

For default-event evaluation, report precision, recall, false-negative and false-positive rates, and the event horizon. Since defaults are rare, accuracy can be misleading. If a separate model estimates probabilities, evaluate calibration and proper scoring rules on held-out observations. A K-Means label is not itself a probability of default.

11 Financial planning and analysis

The framework can support portfolio review, stress testing, concentration analysis, scenario planning, and risk-budget discussions. It should not be treated as a standalone investment instruction. Portfolio exposure depends on price sensitivity, position size, correlation, funding, liquidity, hedges, legal terms, and investor liabilities, not merely the issuer's cluster [17, 16].

A dashboard should report classification and label definitions; currency and horizon; duration and market sensitivities; liquidity measures and data coverage; structural-model uncertainty; centroid distance and stability; evidence for any default flag; and scenario results under interest-rate, inflation, foreign-exchange, fiscal, liquidity, and geopolitical shocks. A label transition should trigger review, not an automatic trade. The classification can identify questions for deeper analysis while independent limits, human review, and scenario-based risk controls remain in force.

12 Law, institutions, and sovereign default

Default is partly an economic event and partly a legally structured concept. Its definition may depend on the bond contract, governing law, grace periods, collective-action clauses, exchange offers, acceleration provisions, and restructuring treatment. A statistical model cannot decide legal default solely from a high yield or cluster centroid. Event data should record source, date, instrument, legal definition, and uncertainty. If definitions differ across instruments or jurisdictions, preserve those distinctions rather than collapsing them silently.

Four propositions must remain separate: an issuer describes its debt as safe; market participants use the debt as a benchmark; regulators assign a particular treatment to the debt; and the claim has specified economic risk characteristics. None entails all the others. Regulatory treatment may be a policy convention; benchmark status may reflect liquidity and market depth; issuer statements are not independent validation; and economic risk depends on the claim's payoff and numeraire [8, 12].

13 Physics, chemistry, and biology: analogies with limits

13.1 Physics and statistical mechanics

A common component resembles a shared field in a multi-agent system, while sovereign-specific deviations resemble local perturbations. Network models can represent connected exposures, and regime changes can be studied as changes in observable distributions. These are methodological analogies. A market is not thereby proven to obey a physical conservation law, and a fitted latent factor should not be called a physical field without a formal mapping.

13.2 Chemistry

Mixture analysis offers an analogy for latent-component decomposition: observed prices may combine multiple influences, and the inverse problem may have multiple solutions. Economic components are not chemical species, however; they can be endogenous, strategic, and institutionally defined. The analogy motivates identification analysis but does not supply it.

13.3 Biology and adaptation

Financial systems adapt as governments change fiscal policy, investors reallocate portfolios, and institutions modify rules. This resembles adaptive systems with feedback. Historical cluster boundaries may consequently drift, making monitoring, retraining criteria, and regime-change diagnostics necessary. The metaphor does not imply that sovereigns are organisms or that policy outcomes are biologically inevitable.

14 Psychology, psychiatry, and behavioural finance

Investors and policymakers may exhibit herding, loss aversion, overconfidence, ambiguity aversion, and reliance on salient ratings or benchmark status. These mechanisms can influence prices and help explain why a common pricing component contains both information and shared beliefs. Behavioural explanations should be tested rather than inferred from market movements alone [11]. Psychiatric diagnoses apply to persons under appropriate clinical standards, not to sovereigns, markets, or institutions. Terms such as "irrational" or "pathological" should not substitute for measurable hypotheses.

15 Linguistics, philosophy, and religion

The words *safe*, *risky*, and *default* are not neutral numerical labels. "Safe" may mean low probability of nominal nonpayment, stable purchasing power, liquidity, collateral usefulness, or low portfolio volatility. A classification scheme must define the term operationally and disclose what it excludes.

The framework distinguishes an observable quantity, a latent construct, an identifying assumption, and an interpretation. An observed yield is a market measurement; a common component is a model-dependent latent construct; a zero-mean normalization is an identifying convention; and "risk-free" is an economic interpretation requiring a stated risk definition. Religious and philosophical traditions can inform normative reflection on promises, obligations, justice, stewardship, and crisis resolution. They do not determine empirical yields or prove a statistical classification. Descriptive claims about risk and normative claims about what governments or creditors ought to do should be reported separately.

16 Political science, geopolitics, diplomacy, and international relations

Repayment capacity is affected by institutional legitimacy, tax administration, political constraints, sanctions, access to international markets, trade dependencies, external security commitments, and diplomatic relationships. Such variables may be relevant features or scenario drivers, but they must be operationalized carefully and supported by evidence.

Diplomatic commitments and mutual-defence arrangements can affect perceived risk in multiple directions: they may reduce some tail risks while creating contingent fiscal liabilities or exposure to conflict. A network of alliances should not be translated mechanically into a low-risk score. The analysis should

distinguish treaty text, implementation, political credibility, fiscal capacity, and actual crisis behaviour. Sovereign debt structures also differ in currency denomination, investor base, central-bank arrangements, external liabilities, capital-market access, and legal jurisdiction. These structural differences may be more informative than a single aggregate rating.

17 Welfare economics and warfare economics

A sovereign-risk classifier can influence borrowing costs, investor access, collateral eligibility, and policy perceptions. Incorrectly classifying a vulnerable state as safe may encourage underpricing of risk; incorrectly classifying a solvent state as risky may intensify funding stress. The possibility that labels influence the system being measured creates a reflexivity problem.

Welfare analysis should ask who benefits and who bears the cost of a classification: taxpayers, pensioners, domestic banks, foreign creditors, public-service recipients, and future generations may be affected differently. The social cost of a false positive need not equal the cost of a false negative. Warfare economics introduces fiscal shocks, mobilisation costs, destruction of productive capacity, trade disruption, sanctions, displacement, and contingent liabilities. These should be modelled as explicit scenarios where relevant. The classifier is intended for financial resilience, public-finance analysis, and risk assessment, not for identifying targets or facilitating violence.

18 Statistics, computer science, AI/ML, and cybersecurity

K-Means is attractive because its objective and centroids are interpretable. Nevertheless, Euclidean geometry assumes that the chosen feature representation is meaningful. Scaling, duplicated features, outliers, or missing-data treatment can alter the clusters. Sensitivity reports should compare plausible specifications rather than presenting one partition as uniquely true. More flexible models, including boosted trees, neural networks, and Bayesian classifiers, should be compared with simple baselines. Complexity is justified only when it yields demonstrable benefits. More expressive models do not resolve exact observational equivalence without added information or assumptions.

Data pipelines should protect against tampering, stale data, unit changes, malicious input, credential compromise, and schema drift. Controls include authenticated source retrieval, documented provenance, versioned raw data, access control and least privilege, secure secret management, checksums where appropriate, schema validation, timestamp checks, unit tests, reproducible environments, versioned models, monitoring for drift, audit logs, and rollback procedures [18]. Cybersecurity improves integrity and reproducibility; it does not guarantee source truth or model validity.

19 Tables for interpretation and governance

Table 1: Table for interpretation.

Concept	**Meaning**	**What it does not establish**
Common component	Shared movement under a specified model	A risk-free rate by definition
Relative sovereign effect	Difference from chosen common reference	Absolute default probability
K-Means cluster	Geometric similarity under chosen features	A natural or legally binding risk class
Safe label	Relatively low measured risk under stated scope	Zero risk in every currency and horizon
Risky label	Elevated measured vulnerability	Certain or imminent default
Default label	Documented event or explicit event criterion	Merely a high yield or distant centroid
Graph regularization	Smoothing informed by a specified network	Identification without added information

Table 2: Table for governance.

Failure mode	Diagnostic	Mitigation
Weak identification	Unstable estimates or weak residualized variation	Alternative restrictions and external data
Scale domination	Clusters change under rescaling	Training-only standardization and robust alternatives
Label leakage	Implausibly strong test performance	Time-based splits and point-in-time snapshots
Rare defaults	High accuracy but poor event recall	Precision/recall and event-based evaluation
Regime drift	Changing centroids and error rates	Monitoring and predeclared retraining policy
Legal ambiguity	Conflicting event definitions	Instrument-level documentation and legal review

20 Research propositions and falsifiability

The framework generates hypotheses, not guaranteed results.

Hypothesis 1. After controlling for currency, maturity, liquidity, and common market movements, issuer-specific pricing deviations contain information about independently measured sovereign vulnerability.

Hypothesis 2. Combining structural estimates with independent fiscal and liquidity indicators produces more stable clusters than a yield-only feature set.

Hypothesis 3. A cluster labelled Risky has elevated subsequent stress-event frequency relative to Safe under a predeclared event definition and horizon.

Hypothesis 4. K-Means alone does not consistently identify legal default events unless default-related information creates a sufficiently distinct feature profile.

Hypothesis 5. Graph-informed features improve out-of-sample stability only when network weights encode relevant relationships without future-information leakage.

Each hypothesis can be rejected. Report uncertainty intervals where feasible, sensitivity to modelling choices, alternative baselines, and unsuccessful specifications. Statistical significance alone is insufficient; effect size and practical relevance matter.

21 Limitations

Bond prices may not uniquely identify a discount curve or a yield decomposition when observations are sparse or cash flows are heterogeneous. A latent common component can include risk premia and shared market frictions. K-Means favours roughly spherical clusters under Euclidean distance and may fit poorly to non-convex or unequal-density groups. Three clusters do not guarantee three meaningful economic states. Default events are rare, heterogeneous, and legally contingent. Geopolitical and institutional variables may be difficult to quantify without subjective judgements. Finally, model outputs can affect behaviour, changing the data-generating process. The appropriate response is to document conditions under which the classification is informative and decline unsupported interpretations.

22 Conclusion

This treatise integrates continuous-time bond valuation, duration, structural identification, econometric decomposition, graph-aware modelling, and three-class K-Means clustering. The central mathematical result is that relative issuer effects can be identified through differences under an additive common-component model, while the absolute common component remains ambiguous without additional restrictions. A statistical normalization can select a decomposition, but cannot prove that the common component is risk-free.

K-Means adds a practical classification layer. It groups sovereigns by a selected feature representation; it does not inherently know the labels Safe, Risky, and Default. Those labels require explicit definitions, independent validation, and special care around documented default events. Safe is relative to the chosen sample, currency, horizon, and risk dimension. The broader disciplinary lesson is that a sovereign bond is simultaneously a financial claim, a legal contract, an institutional object, and a political-economic instrument. Interdisciplinary context can improve feature selection, scenario design, governance, and interpretation, but no analogy substitutes for identification or evidence [14, 15].

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[19] For each sovereign event, consult the applicable bond documentation, governing law, restructuring terms, and jurisdiction-specific primary sources.

Glossary

Common component
: A latent component shared across observations under a specified model.

Default event
: A documented failure to meet a contractual obligation or another explicit default criterion.

Duration
: Sensitivity of bond price to a specified shift in discount rates.

Endogeneity
: Joint determination or correlation of an explanatory variable with unobserved outcome determinants.

Identification
: Conditions under which distinct parameter values imply distinct distributions of observables.

K-Means
: An unsupervised algorithm minimizing within-cluster squared Euclidean distances.

Liquidity risk
: Risk that an asset cannot be traded promptly near its estimated value.

Model leakage
: Use of information unavailable at the relevant prediction time.

Risk-free claim
: A claim free of a specified risk under an explicit currency, horizon, payoff, and convention; not necessarily free of all risks.

Safe label
: A relative empirical category supported by a predeclared definition and validation evidence.

Structural identification
: Recovery of interpretable model components using restrictions and observable variation.

Term premium
: Compensation associated with holding a longer-maturity instrument rather than rolling over shorter-maturity claims, under a given decomposition.

Weak identification
: A condition in which parameters are theoretically identified but estimated imprecisely or unstably.

The End